

**1AC**

### **This contains aprioris**

We'll get to security in rebuttal – otherwise it creates a timeskew in the 1AC as it moots 1+ min and we can't be prepared for all Ks.

## Aprior Spec

Interp: Debaters must specify a delineated text in the 1AC/1NC whether there are A prioris or not.

To clarify, A prioris are independent reasons to affirm/negate

Violation: They don't spec

Norming – Hidden Aprioris ruin the educative value of debate by skirting clash which creates a competitive incentive to exclude debaters with learning disabilities by hiding tricks. Kills fairness via exempted or hidden tricks that moots the 1AC. Spec Solves – It alerts debaters to contest arguments AND disincentivizes nonstarter cheapshots.

Fairness is a voter - debate is a competitive activity that requires fairness for objective evaluation.

Drop the debater to deter future abuse.

Competing interps – [a] reasonability is arbitrary and encourages judge intervention  
[b] it creates a race bottom of questionable argumentation.

No RVIs – [a] illogical, you don't win for proving that you meet the burden of being fair

[b] RVIs incentivize baiting theory and prepping it out which leads to maximally abusive practices

## C1: Amazon---1AC

**Only the plan can save the Amazon from unchecked deforestation.**

**Yunus '25.** Independent researcher from Riau Province, Pursuing a master's degree in biological sciences at Khon Kaen University. "Beijing's green mirage: How China drives environmental destruction abroad while claiming climate leadership at home." *Lowy Institute*. June 10, 2025.

<https://www.lowyinstitute.org/the-interpreter/beijing-s-green-mirage-how-china-drives-environmental-destruction-abroad-while>. arjun

China's role in driving deforestation in Brazil's Amazon Basin is increasingly alarming. The region loses about 133 square kilometres of forest each year, equivalent to over 400 football fields daily, primarily due to agricultural expansion for beef and soybean production. These commodities feed global markets, with China being a dominant consumer. In 2022, 96 per cent of China's soy imports originated from soy-producing regions linked to deforestation, compared to only 55 per cent for the European Union. This stark contrast highlights China's disproportionately large impact on Amazon deforestation. Although Brazil has recently made strides in curbing forest loss through policy reforms, stronger enforcement, and indigenous land protections, this progress is undermined by China's rising demand. It threatens to reverse environmental progress in Brazil.

Indonesia offers a stark, immediate example of ecological damage tied directly to China's regional trade ambitions. While China boasts investments in clean energy and infrastructure through its Belt and Road Initiative, it is simultaneously driving forest destruction via its voracious demand for palm oil, pulpwood, and mining materials. A Traise Earth analysis found palm oil deforestation on Sumatra surged 3.7 times between 2020 and 2022, a spike fuelled in part by soaring Chinese demand and domestic consumption. The study also revealed China has overtaken the EU and India as Indonesia's top palm oil buyer, snatching a growing share of exports. Yet despite corporate promises and sustainability initiatives, enforcement is patchy at best, traceability remains murky, and accountability is all too often absent.

This points to a deeper governance challenge, one that transcends trade balances and national boundaries. The Global Biodiversity Framework and the Paris Agreement are only as robust as their weakest links, and transboundary supply chains are among them. By excluding embedded emissions and biodiversity footprints from their global commitments, nations such as China effectively outsource environmental degradation while taking credit for national progress. It's a loophole substantial enough to undermine the integrity of global environmental governance, and one that allows for the pretence of leadership while the environmental debt is run up somewhere else.

Moreover, the China strategy's moral hazard sets a chilling precedent. If a country with the rich capacities and strong political influence of China can pursue a two-track environmental policy – green at home, grey abroad – why shouldn't others follow suit? In an already stressed world, this dishonesty has the potential to lock us in a vicious cycle of hollow commitments and permanent ecological tipping points. It is greenwashing at a national level.

**Amazon deforestation destroys planetary health— climate change extinction but also diseases comparable to covid.**

**Chies '20.** Scientist at Departamento de Genética, Universidade Federal do Rio Grande do Sul, citing amongst others:

Joel Henrique **Ellwanger**. Brazilian Researcher at the Departamento de Genética, Universidade Federal do Rio Grande do Sul.

Philip M. **Fearnside**. Bachelor's degree in Biology, Colorado College, Master's degree in Zoology, PhD. in

Biological Sciences, University of Michigan.

“Beyond diversity loss and climate change: Impacts of Amazon deforestation on infectious diseases and public health.” *Anais da Academia Brasileira de Ciências*. April 17, 2020.

<https://pubmed.ncbi.nlm.nih.gov/32321030/e>. arjun

The Amazon Basin is the largest river system in the world, encompassing more than 7 million square kilometers distributed between Brazil, Bolivia, Colombia, Ecuador, French Guiana, Guyana, Peru, Suriname, and Venezuela. With enormous biodiversity, most of Amazonia is located in Brazil, representing the largest biome of a country known by its thriving nature (Garda et al. 2010, Peres et al. 2010, Latrubesse et al. 2017, Metzger et al. 2019).

Amazonia is a unique biome in many aspects, with importance in different spheres of life. This biome's huge diversity of animal and plant species is per se a strong justification for its preservation. In addition, many benefits to human life come from direct or indirect interactions with Amazon ecosystems. Indigenous and traditional peoples live and preserve ecosystems and their cultures through the forest. For urbanized communities, among other benefits, the Amazon forest is a source of food, chemical compounds for the development of medicines, and raw materials for a wide variety of industries. The Amazon rainforest is also crucial for maintaining planetary health due to its pivotal role in regulating the Earth's climate. In a broader perspective, protecting Amazon ecosystems is essential for biodiversity preservation, climate regulation, energy production, food and water security; it is also important for pollination, natural/biological control of pests, the region's economy and human health, not forgetting to mention its aesthetic and cultural value. Amazon ecosystems have an important role for the dynamics and control of zoonotic diseases and vector-borne infections, a very important, although sometimes neglected, point which will be discussed in detail throughout this article (Suffredini et al. 2004, Alho 2012, Bieski et al. 2015, Baker & Spracklen 2019, Metzger et al. 2019, Moraes et al. 2019, Valli & Bolzani 2019).

Global warming is an aspect of climate change that has consequences for the spread of human infections (Shuman 2010, Watts et al. 2018). “Climate change” refers to changes in climate properties (temperature, precipitation, extreme events, and wind patterns) that persist for a long period of time (decades or longer) (Liang & Gong 2017). The Earth's average temperature is increasing at least in part due to anthropogenic actions, such as the emission of greenhouse gases from industries and extensive use of fossil fuels (Levitus et al. 2001, Huber & Knutti 2012, Powell 2015, Letcher 2019).

A robust body of evidence shows that deforestation of Amazon forest is a fundamental driver of climate change (Shukla et al. 1990, Werth & Avissar 2002, Malhi et al. 2008, Khanna et al. 2017, Lovejoy & Nobre 2018, Baker & Spracklen 2019). About 20% of the original Amazon forest cover in Brazil has already been deforested (INPE 2019). Recently, policies, laws, agreements, funds, and practical actions focused on Amazon protection have been weakened in Brazil, encouraging deforestation (Carvalho et al. 2019, Ferrante & Fearnside 2019, Pereira et al. 2019, Seymour & Harris 2019). It is evident that, after a period when conservation policies were intensified in Brazil, which resulted in positive impacts on Amazon protection (West et al. 2019), deforestation in the region started to grow again (Artaxo 2019, Fearnside 2019). Along with environmental problems and the weakening of environment-related policies, lack of science funding will also cause economic losses for the country (Magnusson 2019). The current degradation status of the Amazon rainforest is already very serious, but, if this situation is not appropriately handled, the forest and climate situation on Earth will become increasingly worrying.

The association between anthropogenic action in the Amazon rainforest, climate change, alterations in vector dynamics, human migration, genetic changes in pathogens and the poor social and environmental conditions in many Latin-American countries can give rise to the “perfect storm” for the emergence and re-emergence of human infectious diseases in Brazil and other Amazonian countries. The recent Zika virus epidemic and the spread of dengue, chikungunya and yellow fever cases are just a few examples of diseases that affect countries in the Amazon region and even other regions of the globe (Ellwanger & Chies 2016, Lima-Camara 2016, Schuler-Faccini et al. 2016, Donalisio et al. 2017, Goldani 2017, Gregianini et al. 2017). Amazonian fauna hosts a huge diversity of well-known pathogens, as well as many other potential new or even unknown pathogens (Vasconcelos et al. 2001, Abrahão et al. 2010, Dornas et al. 2014, Bonato et al. 2015, Soares et al. 2015, Barros et al. 2018, da Silva et al. 2018, Fernandes et al. 2018, Farikoski et al. 2019, Franco Filho et al. 2019, Medeiros & Vasconcelos 2019). Although several of these pathogens may have low epidemic potential in humans, this abundance of microorganisms in the Amazon region indicates that emergence of new infections from the forest is a constant threat to human health.

The link between “environmental imbalances” and “emerging infectious diseases” is already well established in the literature (Daszak et al. 2001, Weiss & McMichael 2004, Jones et al. 2008). However, descriptions and discussions regarding the factors involved in the emergence of infectious diseases due to deforestation specifically in the Amazon region are still scarce. As an attempt to foster this discussion, we review problems and activities associated with the Amazon deforestation and their impacts on the dynamics of infectious

diseases and on human/public health.

## C2: Hypersonics---1AC

### Defense-dodging Chinese hypersonics are set to arrive next year.

**DSA '25.** Prominent Malaysian news portal covering Asian defense. “Mach 5 Shape-Shifter: China’s New Hypersonic Missile Can Morph Mid-Flight to Evade Any Defence.” *Defence Security Asia*. November 2.

[https://defencesecurityasia.com/en/china-morphing-hypersonic-missile-mach5-pla-weapon/#google\\_vignette](https://defencesecurityasia.com/en/china-morphing-hypersonic-missile-mach5-pla-weapon/#google_vignette). et

(DEFENCE SECURITY ASIA) — In the shadowed corridors of China’s defence laboratories, where Confucian discipline meets aerospace ambition, the People’s Liberation Army (PLA) has unveiled a weapon that defies conventional physics and redefines twenty-first-century deterrence.

A new hypersonic missile prototype capable of “morphing” its aerodynamic shape mid-flight at Mach 5 ( $\approx 6,174$  km/h) has emerged from within the National University of Defence Technology (NUDT) — an institution long regarded as the intellectual crucible of China’s missile programs.

The weapon’s existence, corroborated by a peer-reviewed paper in *Acta Aeronautica et Astronautica Sinica* on 20 October 2025, showcases a technological leap that merges retractable composite wings with adaptive AI-driven flight algorithms. “This is the holy grail of hypersonic flight.”

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the study implies, encapsulating decades of international pursuit to overcome the thermal and aerodynamic extremes that plague velocities above Mach 5. At such speeds, atmospheric friction creates temperatures nearing  $3,000^{\circ}\text{C}$ , hot enough to liquefy conventional alloys — yet NUDT’s prototype survives, evolving mid-course to balance lift, drag, and maneuverability. When its wings are retracted, the missile resembles an arrowhead-shaped spear, slicing through the atmosphere with minimal drag to extend its reach beyond 2,000 kilometres (1,240 miles) — roughly the distance from Hainan Island to the Philippines. When extended, the same wings deliver blistering lift, allowing instantaneous altitude adjustments and tight-angle turns that could evade radar-guided interceptors or terminal-phase anti-missile defences. Each wing section is believed to employ carbon-carbon composites or ultra-high-temperature ceramic matrix materials, operating through precision electromechanical actuators validated via Hardware-in-the-Loop (HIL) simulations — a hybrid testing regime where real components interact with digital aerodynamic models to ensure structural integrity at hypersonic stress levels. The result is a machine that not only flies faster than the eye can track but reshapes itself faster than defenders can predict. From DF-17 to the Morphing Era China’s hypersonic odyssey began in the 1990s, catalysed by U.S. initiatives in prompt global-strike research. By 2019, the PLA Rocket Force (PLARF) introduced the DF-17 hypersonic glide vehicle, which travels at Mach 5–10 and can “skip” across the upper atmosphere to defeat ballistic-missile interceptors. The DF-17 was Beijing’s strategic proof-of-concept — a weapon that demonstrated reach, speed, and unpredictability. Yet its form remained static, a limitation that morphing technology now promises to shatter. Where the DF-17 relies on fixed aerodynamics, the new morphing prototype can dynamically alter its profile in flight, effectively bridging the gap between hypersonic glide vehicles (HGVs) and high-maneuverability cruise missiles. NUDT’s involvement is a signal of systemic integration: the university has already led breakthroughs in scramjet propulsion, plasma stealth reduction, and AI-guided control systems, all of which feed directly into China’s next-generation missile complex. Its laboratories — often compared to the Massachusetts Institute of Technology for missiles — operate under the PLA Strategic Support Force, ensuring that academic innovation flows seamlessly into operational doctrine. In engineering terms, morphing flight represents a culmination of multiple disciplines: computational fluid dynamics, adaptive materials, electromechanical actuation, and machine-learning control loops. Such convergence is what allows the prototype to transition from boost-glide phase to terminal strike without compromising stability — a feat Western programs like the

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U.S. Air Force’s ARRW (AGM-183A) and DARPA’s HAWC have struggled to achieve due to overheating and control-law issues.

With ARRW cancelled in 2023, Beijing has now seized a technological advantage that could endure into the next decade.

Mach 5 and the Metallurgy of Hell

At Mach 5, the sky becomes a furnace, and materials behave less like metals and more like fluids.

The PLA’s morphing missile tackles this inferno through a combination of titanium alloys, ceramic ablatives, and proprietary coatings developed under China’s 14th Five-Year Plan (2021–2025), which earmarks hypersonic research as a “national strategic priority.”

Funding exceeds USD \$100 billion ( $\approx$  RM 480 billion) annually in defence R&D — a budget dwarfing that of most Indo-Pacific nations combined.

Propulsion is believed to rely on a dual-phase system: a solid-fuel booster for the initial ascent and a scramjet or rocket-augmented glide stage for sustained flight.

Such architecture provides flexible launch profiles — from ground-based silos and mobile transporters to air-launched variants that could emerge from H-6N bombers or future H-20 stealth platforms.

The morphing mechanism itself may employ shape-memory alloys and AI-linked actuators that respond within milliseconds to changing airflow conditions, giving the missile near-biological agility.

During the cruise phase, the wings stay folded to minimize drag; as it approaches its target, they extend partially to generate lift, allowing sudden altitude jumps or lateral turns — a nightmare for radar tracking

## algorithms.

Real-time flight adjustments are governed by onboard autonomous navigation software, merging inertial sensors, satellite updates, and AI prediction models to maintain precision even amid GPS jamming or electronic countermeasures.

Studies referenced in the NUDT paper suggest a 30–50 percent increase in maneuverability compared with traditional fixed-geometry HGVs.

In practical terms, that means the missile could out-turn interceptors such as the SM-6, Aegis Ashore, or Japan's new Type-03 Kai SAM — compressing reaction windows for defenders from minutes to seconds.

Redefining A2/AD Across the Indo-Pacific

The strategic implications are profound.

For Taiwan — barely 180 kilometres from mainland China — the morphing missile adds a new layer to Beijing's anti-access/area-denial (A2/AD) umbrella.

Deployed alongside the YJ-21 anti-ship hypersonic missile and the DF-26 intermediate-range “Guam Killer”, it could saturate regional defences in a multi-axis attack. From the air, sea, or land, waves of hypersonic projectiles could overwhelm even advanced U.S. and Japanese systems, striking carrier strike groups, command centres, or runway complexes before detection.

Stealth aircraft such as the F-35 Lightning II and B-21 Raider, designed for low observability, may struggle against a missile capable of adjusting its radar cross-section mid-flight — retracting surfaces to reduce RCS during approach and extending for maneuver only in terminal phases.

That renders traditional “predict-and-intercept” tactics obsolete.

For ASEAN nations, particularly those bordering the South China Sea, the weapon shifts regional deterrence.

It could be integrated into coastal batteries across Hainan, the Paracels, or Fiery Cross Reef, giving Beijing the capacity to strike any naval presence — Vietnamese, Filipino, or Malaysian — within minutes.

Analysts warn that a morphing hypersonic system deployed on PLAN cruisers like the Type 055 Renhai-class or future aircraft carriers such as the Fujian could enforce the “nine-dash-line” through deterrence by punishment, not just presence.

The Indo-Pacific Command would be forced to re-evaluate its defence architecture, accelerating programs like the Aegis Baseline 10, THAAD 2.0, and the Glide Phase Interceptor (GPI) to restore strategic balance.

According to U.S. DoD assessments in 2024, China's DF-17 deployments already “transform the PLA's missile force.” The morphing variant pushes that transformation into uncharted territory — a realm where speed meets shape and strategy becomes fluid.

Global Arms Race and the Hypersonic Horizon

Worldwide, only a handful of nations are pursuing morphing aerodynamics in hypersonics.

Russia's Avangard HGV and Kh-47M2 Kinzhal demonstrate raw speed but lack variable geometry, limiting their agility.

The U.S. remains active through DARPA's Hypersonic Air-breathing Weapon Concept (HAWC) and the Hypersonic Attack Cruise Missile (HACM), yet multiple failures and funding constraints have delayed operational readiness to the late 2020s.

In contrast, Beijing's state-driven ecosystem — anchored by military-civil fusion and unlimited budget support — allows simultaneous advances in aerothermodynamics, plasma control, and AI-guided flight mechanics.

Facilities like the JF-22 and JF-12 wind tunnels simulate conditions up to Mach 30, giving Chinese scientists a testing environment that no Western nation currently matches.

With flight trials expected by 2027, the PLA could achieve operational capability well before NATO's counter-systems mature.

Beyond China's borders, the proliferation risk is real.

Through its Belt and Road Initiative, Beijing may eventually export simplified variants to strategic partners like Pakistan, Iran, or Saudi Arabia, mirroring Turkey's UAV strategy that turned the Bayraktar TB2 into a global brand.

Such a move would introduce hypersonic strike capabilities to regions previously limited to subsonic cruise missiles, dramatically compressing deterrence stability in the Middle East and South Asia.

**The plan solves---it decks China's supply of niobium, which is uniquely key.** **Torres 24.** Chief Operating Officer, Irregular Warfare Initiative. "Hypersonic Hegemony: Niobium and the Western Hemisphere's Role in the U.S.-China Power Struggle." CSIS. March 4, 2024. <https://www.csis.org/analysis/hypersonic-hegemony-niobium-and-western-hemispheres-role-us-china-power-struggle>. et

Introduction In the evolving landscape of global defense, the arms race has metamorphosed from a contest of nuclear might to one of unparalleled speed. Hypersonic weapons (capable of exceeding five times the speed of sound), promise to revolutionize modern warfare.

China's strides in the hypersonic field are a manifestation of its broader strategic intentions and underscore its drive toward technological and military preeminence. This journey toward mastering hypersonic technology is not merely for show; it is about redefining the global balance of military power.

A successful deployment of these weapons would enable China to redefine the term "first-strike advantage." Such a capability is not merely about striking first; it is about striking in a manner that leaves the opponent minimal time or capacity to react, effectively nullifying their defensive postures. This introduces a dangerous paradigm wherein the traditional cushion of time provided by early warning systems is drastically reduced. For the United States, this might mean that even with the world's most advanced detection systems, the window to act could be so minimal that it might be rendered ineffective. As the United States and China jostle for dominance in this arena, the strategic significance of an elemental material, niobium, emerges as a pivotal concern, and with it, China's rising dominance in the Western Hemisphere's mining sector.

### **Niobium: The Aerospace Marvel**

Vacuum-grade niobium's role in aerospace is not a newfound revelation. Its unparalleled resilience against extreme thermal stresses, withstanding temperatures over 2,400 degrees Celsius, renders it indispensable for critical components in hypersonic vehicles. Beyond its inherent properties, niobium's pivotal role lies in its use for crafting heat-resistant superalloys essential for hypersonic missiles and the broader aerospace sector. Its low density compared to other refractory metals contributes to a high strength-to-weight ratio, which is essential for reducing the weight of aerospace components. This reduction in weight directly impacts fuel efficiency and payload capacity, two critical factors in aerospace design.

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For example, companies like SpaceX and Hermeus depend on niobium C103 for their spacecrafts, which require extremely high temperatures that surpass that of other superalloys. For decades, niobium has played a pivotal role in the U.S. aerospace industry, with its notable use in the innovative designs of the iconic Gemini and Apollo programs of the 1960s and 70s. However, despite its significance, the United States depends entirely on niobium imports, with no substantial domestic mining since 1959. This reliance introduces a severe risk to its supply chain. Of the estimated 8,800 metric tons imported annually in 2022, a significant majority comes from Brazil (66 percent) and Canada (25 percent). This heavy reliance on just two primary sources—both neighbors of the United States in the Western Hemisphere—exposes the United States to considerable national security and economic vulnerabilities. The situation becomes even more precarious considering China's dominant position in the niobium sector and its growing footprint in the hemisphere. China's Stake in Brazil's Mineral Monopoly Brazil dominated global niobium production in 2020, accounting for over 91 percent of production. With reserves estimated at 842 million metric tons, Brazil produces roughly 120,000 metric tons annually. Brazil's high production comes from large mines such as Araxá and Catalão, with Companhia Brasileira de Metalurgia e Mineração (CBMM) controlling 75 percent of Brazil's output. China has recognized the potential of niobium for over a decade. In 2011, a consortium of five Chinese companies acquired a 15 percent stake in CBMM. This engagement intensified in 2016 when China Molybdenum Co. Ltd. (now known only as CMO) secured ownership of the Chapadão and Boa Vista mines, further strengthening China's position in the niobium market. The importance of niobium was further highlighted in the Brazilian political arena in 2018. Then presidential candidate Jair Bolsonaro emphasized niobium's role in Brazil's "economic independence." Despite Bolsonaro's campaign rhetoric focusing on safeguarding this critical commodity from foreign control and advocating for its national governance, Chinese influence in the Brazilian niobium sector continued to grow. By 2020, Chinese entities controlled approximately 26 percent of Brazil's niobium production. This control not only ensures China's preferential access and influence over pricing dynamics in the niobium supply chain, but also positions it advantageously in a global context. China managed to maintain and even strengthen its position at the subnational level under Bolsonaro. CMO for instance provided \$1.2 million in Covid-19 aid to the city of Catalão, demonstrating China's strategic engagement beyond mere commercial interests. China's

influence over Brazil's niobium production conforms to a pattern of growing ownership and sway over the regional mining industry, a trend with substantial environmental, political, and security implications. Such tactics could force nations into making diplomatic compromises, ceding trade advantages, or grappling with economic dilemmas, thereby solidifying China's geopolitical standing. The United States is not immune to this exposure; in 2022 the U.S. Geological Survey identified niobium as the second most critical of 50 minerals, falling behind only gallium in its criticality to U.S. national security and economic growth. Defense Implications China's hypersonic resolve has been remarkable. By 2018, it had conducted over 20 times as many tests as the United States. According to the Pentagon, the United States is still lagging. This hypersonic prowess, combined with China's stranglehold on niobium, places the United States in a perilous position.

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The strategic importance of niobium in next-generation defense systems cannot be overstated. As the U.S. military and its defense contractors increasingly rely on niobium-based superalloys to produce a wide range of equipment, from aircraft components to hypersonic missile systems, any disruption in the niobium supply chain could have significant repercussions.

Overall, China's growing influence and control over critical mineral supply chains poses a distinct challenge. Under the Biden administration, the United States and the European Union placed export controls and restrictions on strategic and critical minerals to curb China's dominance in artificial intelligence and semiconductors. In retaliation, China imposed their own limitations on gallium, germanium, and graphite throughout 2023. A recent analysis by CSIS highlighted that China controls 90 percent of global gallium supplies, 90 percent of graphite, and 60 percent of germanium, all critical to the production of chips and electric vehicle batteries. The critical mineral supply chain has arrived at the forefront of strategic competition between the West and the People's Republic of China.

China's grip on the production, distribution, and pricing of niobium presents another layer of complexity: manipulating niobium's availability to other nations. For the United States, already grappling with the challenges of overdependence on external sources for critical minerals, such a disruption could translate into significant production delays. The consequences could be serious: slower production of critical defense equipment, increased costs due to the potential need for alternative materials, and a cascading effect on existing machinery's maintenance and upgrade cycles. In this highly complex environment where timely responses to emerging threats are vital, these delays could hinder the United States' ability to promptly deploy or develop necessary defense systems.

## **Brazilian mining is the lynchpin.**

Maria **Zupello 25**. Masters from Universita Gregoriana, Contributor at the Diálogo Americas. "China Tightens Control of Critical Minerals in Peru and Brazil." *Diálogo Américas*. May 13, 2025.

<https://dialogo-americas.com/articles/china-tightens-control-of-critical-minerals-in-peru-and-brazil/>.  
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Driven by its ambition to achieve greater control over key raw materials used in high-tech industries for its own gain and exert geopolitical influence, China has been accelerating its expansion in the critical minerals sector worldwide.

The strategy for monopoly over minerals essential for warfare technologies and military modernization has led Beijing to increase its investment and influence in Latin America, a move experts say poses global and national security concerns, as China could use critical minerals as bargaining chip in conflicts or disrupt supply chains to hinder technological advancements.

Critical minerals, such as lithium, copper, cobalt, nickel, niobium, and rare earths, are concentrated in a few regions for which South America has a prominent role. Beyond the lithium triangle, other countries in the region such as Brazil and Peru offer great opportunities for the hunt for critical minerals.

According to the independent research center Instituto Igarapé, based in Rio de Janeiro, Brazil has 94 percent of the world's niobium reserves, 22 percent of graphite, 16 percent of nickel, and 17 percent of rare earths, the third largest deposit on the planet. It is worth noting that Niobium's strategic role in advanced defense is paramount. Niobium superalloys are used in an array of advanced weaponry, from aircraft components to hypersonic missiles.

According to data from Peru's Ministry of Energy and Mines, with 110 million tons of reserves, this Andean country is the third largest producer of copper in the world, essential for electromobility, renewable energy, and electrical infrastructure. Significant lithium deposits have also been discovered in Puno, in the south of the country, which are essential for batteries. In addition, Peru is recognized as the second largest producer of zinc in the world, surpassed only by China.

In the last six months, China's investments in the critical minerals sector in Brazil and Peru have stepped up.

Beijing invested some \$800 million in Brazil through a series of operations that have raised the alarm about the risk of monopoly.

In November 2024, state-owned China Nonferrous Metal Mining (CNMC) bought Peru's Minsur S.A.'s Brazilian subsidiary Mineração Taboca, which operates the Pitinga mine, for \$340 million. The mine, located in Presidente Figueiredo, in the heart of the Amazon, 100 kilometers from Manaus, contains huge reserves of tin, from which niobium, tantalum, and hafnium are also

obtained.

"The purchase of Taboca reflects China's growing presence in Latin America, which has become China's playground. Brazil runs the risk of becoming a mere chessboard in a technological war, losing control of strategic sectors," independent analyst Cezar Roedel told Brazilian daily Gazeta do Povo.

Last February, the British multinational AngloAmerican announced an agreement to sell the Barro Alto and Niquelândia ferronickel mines and processing plants in the west-central state of Goiás to a subsidiary of Chinese company MMG, for some \$500 million. In total, these facilities produced 39,400 tons of nickel in 2024. The agreement also includes two future projects for the production of high-quality nickel: the Jacaré deposits in Pará state and Morro Sem Boné in Mato Grosso state, with a potential of some 300 million and 65 million tons, respectively. In addition, in March, Chinese company Baiyin Nonferrous Group Co. (BNMC) bought Mineração Vale Verde (MVV) and its copper plant in Alagoas state, in northeastern Brazil.

Chinese car manufacturer BYD also confirmed that it has acquired the exploration rights to two lithium reserves, essential for electric car batteries, in the Jequitinhonha Valley in the state of Minas Gerais. This region is home to the country's main known lithium reserves and is already being exploited by Brazilian companies Sigma Lithium and Companhia Brasileira de Lítio.

"All these operations clearly show that China in Latin America is more interested in extracting critical minerals than in creating supply chains,"

### **Chinese hypersonic confidence escalates.**

Richard **Weitz 21**. M. Phil. in Politics from Oxford University, M.Sc. in International Relations from London School of Economics, BA, Harvard College. "China's Hypersonic Missiles: Methods and Motives." *Jamestown*. July 30, 2021.

<https://jamestown.org/chinas-hypersonic-missiles-methods-and-motives/>.

The PLA will likely employ hypersonic systems in combination with its subsonic and supersonic delivery systems. Because of their speed and unique trajectory capabilities, hypersonic missiles can, as first-strike weapons, facilitate follow-on attacks by non-hypersonic strike systems by disabling an adversary's air and missile defense systems. One PRC defense expert specifically observed that the range of the DF-17 would enable the system to reach the Terminal High Altitude Area Defense (THAAD) BMD system in South Korea and the SM-3 BMD interceptor in Japan, "which are security threats to China" (Global Times, October 1, 2019).

In addition to directly bolstering the PLA's warfighting capabilities, China seeks hypersonic delivery systems to weaken Washington's extended deterrence guarantees to its allies and partners in Asia. A recent report by the U.S.-China Economic and Security Review Commission notes that, "Capabilities that can credibly threaten the U.S. military also support Beijing's aim to intimidate and coerce regional states by fueling doubts about U.S. ability or willingness to intervene in a crisis."[5][5] Jacob Stokes, "China's Missile Program and U.S. Withdrawal from the Intermediate-Range Nuclear Forces (INF) Treaty," U.S.-China Economic and Security Review Commission,...

#### **Conclusion**

The PLA's hypersonic delivery systems intensify Beijing's challenge to the U.S.' political-military primacy in the Indo-Pacific region. The effectiveness of China's hypersonic capabilities in battle is uncertain; in addition to the PLA being largely untested in combat, hypersonics are still an emerging military technology. Furthermore, we do not know whether China, like the United States, will choose to only arm its hypersonic missiles with conventional munitions, or whether Beijing will follow Moscow's lead and load some of its hypersonic delivery systems with nuclear warheads. In any case, China's novel hypersonic capabilities could, alongside the growing power of the PLA in general, make PRC decision-makers more confident about their ability to employ force successfully against the U.S. military. If Chinese leaders believe that they could employ these capabilities to preemptively destroy or disable key U.S. defense systems, then this would give Beijing greater incentive to strike first, raising the risk of escalation and war in a crisis.[6] If other Asian countries believed that China could more effectively fight U.S. forces, this could weaken the credibility of U.S. security guarantees to allies in Asia and further undermine regional stability.

Some of the measures that the United States and its Indo-Pacific allies are already taking to minimize damage from the PLA's traditional missile strike capability will also enhance defense against hypersonic weapons, for example passive measures that the U.S. military is pursuing to decrease the vulnerability of U.S. forward-based forces and facilities to China's ballistic and cruise missiles that include deception, dispersal,

hardening, concealment, and mobility, as well as redundancy, recovery, and reconstitution.[7] In the future, more active (and expensive) responses could encompass disrupting hypersonic data links and sensors, space-based sensors that can track missiles in the upper atmosphere, and novel technologies for interceptors. The 2019 Missile Defense Review directs that, “Moving forward, the United States, allies, and partners will pursue a comprehensive missile defense strategy that will deliver integrated and effective capabilities to counter ballistic, cruise, and hypersonic missile threats” (Office of the Secretary of Defense, January 17, 2019). But U.S. policymakers have yet to agree on the balance of funding for hypersonic weapons systems, enabling technologies, supporting research and development infrastructure, and hypersonic missile defense (CRS, updated July 9).

### C3: Aviation---1AC

**China is closing the air superiority gap, but their approach depends on mass production---2026 will be key.**

**Phillips 25** [Morgan Phillips, Reporter @ FoxNews; 11-13-2025; "Skies at stake: Inside the US-China race for air dominance"; Fox News;  
<https://www.foxnews.com/politics/skies-stake-inside-u-s-china-race-air-dominance>; DOA: 12-21-2025]  
shaan

China's leap forward

**China's airpower modernization has accelerated as the U.S. reshapes its force.** Beijing has zeroed in on three priorities – stealth, engines and carriers – the areas that long held its military back.

The Chengdu J-20, China's flagship stealth fighter, is being fitted with the new WS-15 engine, a home-built powerplant meant to rival U.S. engines.

"It took them a while to get out of the blocks on fifth generation, especially to get performance anywhere near where U.S. fifth gen was," Heginbotham said. "The J-20 really does not have a lot of the performance features that even the F-22 does, and we've had the F-22 for a long time."

Meanwhile, China's third aircraft carrier, the Fujian, was commissioned this fall – the first with electromagnetic catapults similar to U.S. Ford-class carriers. The move signals Beijing's ambition to launch stealth jets from sea and project power well beyond its coast.

Together, the J-20, the carrier-based J-35, and the Fujian give China a layered airpower network – stealth jets on land and at sea backed by growing missile coverage.

Chinese military writings identify airfields as critical vulnerabilities. PLA campaign manuals call for striking runways early in a conflict to paralyze enemy air operations before they can begin. Analysts believe a few days of concentrated missile fire could cripple U.S. bases across Japan, Okinawa and Guam.

"The U.S. bases that are forward deployed – particularly on Okinawa, but also on the Japanese mainland and on Guam – are exposed to Chinese missile attack," said Mark Cancian, a retired Marine colonel and senior advisor at the Center for Strategic and International Studies. "In our war games, the Chinese would periodically sweep these air bases with missiles and destroy dozens, in some cases even hundreds, of U.S. aircraft."

Heginbotham said that missile-heavy strategy grew directly out of China's early airpower weakness.

"They didn't think that they could gain air superiority in a straight-up air-to-air fight," he said. "So you need another way to get missiles out – and that another way is by building a lot of ground launchers."

Different strategies, same goal

The two militaries are taking different paths to the same target: air dominance over the Pacific.

The U.S. approach relies on smaller numbers of highly advanced aircraft linked by sensors and artificial intelligence. The goal: strike first, from long range, and survive in contested skies.

**China's model depends on volume – mass-producing fighters, missiles, and carrier sorties to overwhelm U.S. defenses and logistics.**

**U.S. fighter aircraft – F-35s, F-15s, F-22s – are relatively short-legged, so they have to get close to Taiwan if they're going to be part of the fight," Cancian said. "They can't fight from Guam, and they certainly can't fight from further away. So if they're going to fight, they have to be inside that Chinese defensive bubble."**

Both sides face the same challenge: surviving inside that bubble. **China's expanding missile range is pushing U.S. aircraft farther from the fight**, while American bombers and drones are designed to break back in.

The fight to survive

Heginbotham said survivability – not dogfighting – will define the next decade of air competition.

"We keep talking about aircraft as if it's going to be like World War II – they go up, they fight each other. That's not really our problem," he said. "Our problem is the air bases themselves and the fact that aircraft can be destroyed on the air base."

China, he warned, is preparing for that reality while the U.S. is not.

"They practice runway strikes in exercises, they're modeling this stuff constantly," Heginbotham said. "Unlike the United States, China is hardening its air bases. The U.S. is criminally negligent in its refusal to harden its air bases."

Cancian's war-game findings echo that vulnerability. He said U.S. surface ships and aircraft would likely have to fall back under missile fire in the opening days of a conflict.

"At the initial stages of a conflict, China would have a distinct advantage," Cancian said. "Now, over time, the U.S. would be able to reinforce its forces, and that would change."

Looking ahead

The Pentagon's fiscal 2026–27 budget will determine how fast the U.S. can build out its F-47s, B-21s and CCAs – systems that will shape American airpower through the 2030s.

China's rapid modernization is closing what was once a wide gap, but the U.S. still holds advantages in stealth integration, combat experience and autonomous systems.

"The ability to protect our aircraft, whatever form those aircraft take, on the ground is going to be central to our ability to fight in the Asia theater," Heginbotham said.

"Survivability is going to be key... The ability to protect and disperse your firepower is going to be central to whether we can really stay in this game."

For decades, U.S. air dominance was taken for granted. In the Pacific, that advantage is no longer guaranteed

**Certain minerals are the most important.**

**Brown 25** [Peter J., Senior Fellow for Western Hemisphere Security and Maritime Affairs at the America First Policy Institute and the Vice Chair of the Critical Minerals Leadership Roundtable, presented by the Responsible Battery Coalition ; 12-18-2025; "Our Military Runs on Minerals: Secure Them Before China Seizes Them"; Defense Post; <https://thedefensepost.com/2025/12/18/us-minerals-national-security/>; DOA: 12-21-2025] shaan

If the United States does not act quickly, China's long-term strategy to control these minerals will put the world's most important materials in the hands of the Chinese Communist Party and enable them to hold us hostage.

President Donald Trump and leaders in Washington need to move fast to secure these minerals before it is too late.

Strategic Importance of Minerals

Not all minerals are created equal. The next century of both economic and military competition will be shaped by a handful of materials, including copper, iron, lithium, cobalt, nickel, lead, antimony, and others.

Without reliable access to these minerals, the US economy — and the US military — would be paralyzed.

Ensuring energy, storage, and operational capability in new technologies requires securing these critical minerals now.

Lessons From History

History offers clear lessons.

In the eighteenth century, Britain achieved naval dominance partly through government incentives that spurred the development of the chronometer for precise navigation.

In the nineteenth century, the United States encouraged private citizens to claim guano islands, securing fertilizer needed to feed the nation and strengthen the US merchant marine.

Adversaries, and potential adversaries, do this, too. For example, during World War I, Germany expanded synthetic ammonia production through the Haber-Bosch process after losing access to Chilean nitrates critical for explosives.

China has applied similar logic over the past 15 years. Under President Xi Jinping, Beijing has pursued a deliberate strategy to dominate critical mineral supply chains. China is blessed with extensive deposits of iron ore, copper, and antimony. Where it lacks natural reserves, it has secured them through debt diplomacy, most famously Congolese cobalt, and by dominating processing capacity.

### **China relies on international extraction for those.**

**Gulley 18** [Andrew L., PhD in Mineral Economics; 2018 "Assessing the U.S.-China Competition for Minerals Crucial to the"; No Publication;

<https://scei.fsi.stanford.edu/china-briefs/assessing-us-china-competition-minerals-crucial-development-emerging-technologies>; DOA: 12-21-2025] shaan

China's mineral reliance. China is highly import-reliant on 8 of the 42 minerals that the U.S. is not import-reliant on. Of these 8 minerals, iron ore and mined copper are vital for China's **continuing industrialization and urbanization**.

However, their import reliance reflects unprecedented growth, rather than a lack of domestic resources. In fact, China is the third largest producer of iron ore and second largest producer of mined copper.

All 8 minerals that China is import-reliant on are widely available except beryllium (crucial for military, aerospace, and nuclear applications), for which the U.S. is the dominant global supplier. Because China can obtain the majority of what it needs (except beryllium) from several other countries, its supply risk is relatively lower. Moreover, China appears to have mitigated its import reliance on key minerals: China's state-owned firms have addressed prominent cobalt-mine and copper-mine supply risks through so-called "**infrastructure-for-minerals**" deals with African governments, while China's one-time vulnerability to gallium has likely already been eliminated through domestic capacity expansions.

### **China dominance decks U.S. capabilities---now is the brink.**

**Persinos 25** [John, Author at Avionics International; 2025; "China's Rare Earth Retaliation Threatens the Future of Avionics"; Avionics International;

<https://www.aviationtoday.com/2025/05/01/chinas-rare-earth-retaliation-threatens-the-future-of-avionics/>; DOA: 12-21-2025] shaan

When President Donald Trump launched a trade war with China and imposed sweeping tariffs on Chinese goods, few anticipated the devastating counterpunch that would come from Beijing.

In an aggressive move in mid-April, China—home to more than 80% of the world's rare earth mineral refining capacity—announced restrictions on the export of key rare earths.

While the move was initially framed as economic retaliation, its shockwaves are now reverberating through one of the world's most sensitive and technologically advanced sectors: aviation, and more specifically, avionics.

The Rare Earths That Power the SkiesEmpty heading

Rare earth elements such as neodymium, dysprosium, terbium, and yttrium might sound obscure, but they are essential to modern avionics. These minerals are used in high-performance magnets, sensors, actuators, gyroscopes, and radar systems that form the backbone of a plane's electronic brain.

From inertial navigation systems (INS) to heads-up displays (HUDs) and flight management systems (FMS), avionics simply cannot function without a steady supply of these critical materials.

And therein lies the problem: China dominates the global supply chain—not just in mining, but more importantly in refining. Western countries may have rare earth reserves underground, but without the sophisticated processing infrastructure that China has built up over decades, raw ore is useless.

Avionics in the CrosshairsEmpty heading

The aviation industry is especially vulnerable to supply shocks because of the stringent certification, reliability, and performance requirements that avionics systems must meet. Unlike consumer electronics, you can't simply switch to a lower-grade magnet or cheaper alloy when a key input is in short supply.

Flight-critical systems depend on rare earth-derived components for their precision and durability under extreme conditions, from high G-forces to wide temperature swings.

Flight control computers, GPS receivers, and communication systems all use rare earth-enabled technologies. As China throttles exports, Western avionics manufacturers like Honeywell, Collins Aerospace, and Thales could face production delays, rising costs, and even contract penalties if they can't deliver on time.

Already, avionics industry insiders are warning of bottlenecks in magnetron and gyro production, which are technologies vital for both civilian aircraft and military platforms.

Military and Defense ImplicationsEmpty heading

The implications for defense aviation are even more severe. Fighter jets, drones, and radar systems rely heavily on rare earths, not only for avionics but also for stealth coatings, electronic warfare, and propulsion systems. If the current restrictions escalate into a full embargo, national security could be compromised.

The Pentagon has flagged rare earth dependency as a strategic vulnerability, and Trump's trade policy has spotlighted just how exposed the West really is.

While short-term solutions like stockpiling or sourcing from alternative suppliers (such as Australia or Canada) may offer some relief, they won't solve the core problem: the West is decades behind in rare earth processing and refining.

Policy makers, manufacturers, and defense officials are now considering substantial investment in domestic refining capacity and synthetic alternatives. Without these, the avionics sector risks stagnation, cost inflation, and technological backsliding.

Trump's tariffs may have been intended to pressure China, but the aviation world is ending up as collateral damage. As China tightens its grip on rare earths, avionics developers are finding themselves caught in a geopolitical tailspin, one that threatens innovation, production, and even the safety of future flight.

**Air superiority stops every conflict from escalating.**

**Birkey '24** [Douglas; June 13; Executive Director for the Mitchell Institute for Aerospace Power Studies, Master's in International Relations from Georgetown University; Air and Space Forces, "Past is Prologue: Learn from History and Invest in Combat Airpower,"

<https://www.airandspaceforces.com/opinion-past-is-prologue-invest-in-combat-airpower/>]

As world leaders gathered at Normandy last week to mark the 80th anniversary of the D-Day invasion, it is important to recall that America was not ready for that war. It took two and a half years to generate the combat power necessary to invade Europe, plus another year to finally defeat Nazi Germany. Victory was far from certain during key phases of the conflict.

Today, America is once again inadequately postured to confront burgeoning national security challenges. China is building up forces aggressively in the Pacific; Russia is making gains in its war on Ukraine; Iran is pursuing nuclear ambitions and proxy wars; North Korea is threatening nuclear war; and non-state actors continue to destabilize key regions. However, post-Cold War budget cuts, two decades of strategic distraction in Afghanistan and Iraq, and a succession of aggressive budget cuts have yielded a set of U.S. military capabilities and capacity that fall short of what the threat environment demands. This is especially true for Air Force fighters and bombers—two foundational elements of U.S. national security.

The ability to achieve air superiority when and where it is needed is an essential condition for any successful military operation, regardless of domain. That is why Air Force fighters are so important.

Likewise, a commander's ability to take the fight to the enemy depends on America's long-range attack capability—which is predominantly an Air Force bomber mission. They strike targets deep behind enemy lines, disable command-and-control networks, eliminate production centers, and damage materiel, logistics, and enemy combat forces. Bombers, in addition to submarines and land-based ICBMs, also comprise one leg of America's nuclear triad.

Combined, air superiority and long-range attack deter adversaries in times of peace and yield war-winning effects during conflict.

Both air superiority and long-range attack missions are established military imperatives. In World War II, allied forces would not have made it past the landing beaches had they been under attack from the sky. Long-range strikes played a crucial role throughout the war, degrading the German war machine with debilitating attacks against petroleum refineries, production facilities, rail lines, and more. The reason the war in Europe ended just eleven months after D-Day was that air power had already inflicted so much damage over the years, that ground forces were fighting an already degraded opponent.

The same was true in Operation Desert Storm in 1991. A monthlong air campaign set conditions such that ground forces needed just 96 hours to bring Saddam Hussein to his knees.

Today, in Ukraine, neither side controls the sky, and neither can mount a strategic air attack campaign. As a result, fighting is bogged down in modern-day World War I-style trench warfare. This is the price nations pay if they cannot control the sky and use that dominance to project decisive strikes. Surface forces slaughter each other for incremental daily gains measured in meters.

Following such a course is a recipe for disaster. Yet given the precarious state of U.S. airpower today, it could be what is in store if the U.S. faces a major power, especially China, in combat in the future. Lacking major investment today, commanders will lack the capacity and capability they will need to fight a peer foe.

The decay began after the fall of the Soviet Union in 1991, when the Pentagon cut Air Force combat capacity by half. In continued over the decades that followed, as fighter and bomber inventories shrank and nonstop operational demand in the Middle East, the Balkans, and elsewhere burned out remaining aircraft. At the same time, key fighter and bomber modernization programs were cut short. America built less than half the F-22s required, halted B-2 bomber production at 21, slowed F-35 production, and flew aircraft into their sixth decade and beyond. Fiscal caps imposed by the 2011 Budget Control Act strained Air Force fighters and bombers to the core.

Air Force leaders sounded the alarm six years ago, declaring that “The Air Force is too small for what the nation is asking us to do.” Yet still the cuts continued.

Now, with threats on the rise around the globe, the nation faces a choice: modernize or risk the nation's future security. Said more bluntly, America may lose its next major war it fights for want of capable, properly sized combat airpower.

## **C4: Afghanistan---1AC**

### **Chinese REE extraction is the crux of the Taliban's regime security.**

Muhammad Waqar Khan '25. Master of Engineering in the area of Telecommunication Engineering, Post Graduate Diploma in Computer and Information Sciences, Master in Computer Sciences from Karachi University, Ph.D. Scholar, Machine Learning, Hamdard University. "The geopolitical implications of rare earth minerals extraction in Afghanistan." *Journal of Trace Elements and Minerals*. December 2025. <https://www.sciencedirect.com/science/article/pii/S2773050625000576>. arjun

Background: Afghanistan's Rare Earth Elements (REEs) reserves have become a central point of geopolitical competition following the Taliban's return to power in 2021. This study investigates the complex interplay between China's growing dominance in Afghanistan's mining sector, the failure of Western containment approaches, and the Taliban's unconventional governance of these critical resources. Drawing on satellite imagery analysis, blockchain tracking of cryptocurrency transactions, and interviews with industry experts, the research uncovers how China has secured preferential access to REEs deposits through a "security-for-minerals" bargain with the Taliban. The study documents the Taliban's paradoxical approach to resource governance, combining bureaucratic pragmatism with ideological constraints, resulting in revenue capture (15 % local allocations) and notable leakage (42 % lost to corruption). Findings highlight the emergence of REEs smuggling as a key funding source for regional militant groups, with an estimated \$19 million annually flowing to ISIS-K Khorasan. The research contributes to theoretical debates on the "resource curse" in Islamic governance contexts and the evolving effectiveness of sanctions in an era of digital financialization. Practical implications include recommendations for tiered sanctions regimes and regional monitoring mechanisms to curb conflict financing while maintaining pressure for responsible mining practices. This study offers critical insights into how fragile states' mineral wealth reshapes 21st-century geopolitical competition.

#### 1. Introduction

The global transition to renewable energy and advanced technology has triggered an unprecedented scramble for rare earth elements (REEs) – those seventeen chemically similar minerals including neodymium, dysprosium, and lanthanum essential for everything from electric vehicle batteries to precision-guided munitions. As global demand for REEs surges, control over their supply chains is reshaping international alliances as well as national economy and security policies [1]. As nations race to secure these critical resources, Afghanistan has emerged as an unlikely but pivotal player in this new great game of mineral geopolitics. Afghanistan is considered as Saudi Arabia of rare earth elements specifically lithium. Afghanistan's mineral resources, which include lithium deposits that rival Bolivia's "lithium triangle," are valued between \$1 and \$3 trillion by the U.S. Geological Survey [2,3]. Yet this potential bounty remains largely untapped, caught between competing geopolitical interests, a pariah regime, and the persistent specter of conflict. This study examines Afghanistan's rare earth minerals as both a strategic prize and a potential curse, offering new insights into resource politics in fragile states and the evolving dynamics of global mineral competition. The importance of this research stems from three converging global trends. First, the accelerating energy transition has dramatically increased demand for REEs, with global consumption projected to grow 300–1000 % by 2040 depending on the mineral [4]. Second, growing U.S.-China tensions have turned rare earth supply chains into a national security issue, particularly given China's current dominance (controlling 80 % of global REEs production and 90 % of processing capacity). Fig. 1 shows the global distribution of rare earth reserve and the status of advanced projects.

<<<Figure 1 Omitted>>>

Fig. 1. Global distribution of advanced REEs projects and future demand & supply.

Source: One Earth, 2022.

Third, the Taliban's return to power in Afghanistan has created an unprecedented situation where one of the world's most significant untapped REEs deposits lies under the control of an internationally isolated, theocratic regime. This confluence of factors makes Afghanistan's rare earth resources a critical case study in 21st-century resource geopolitics. Historical context reveals why Afghanistan's mineral potential remains unrealized. Soviet geologist's expert first identified substantial deposits in the 1980s, but decades of conflict prevented development [5]. The post-2001 period saw renewed interest, with the U.S. Department of Defense conducting extensive surveys that confirmed world-class deposits, including the Khanneshin carbonatite complex in Helmand province (containing significant lanthanum and cerium) and lithium resources comparable to Bolivia's "lithium triangle" [6]. However,

corruption, insecurity, and infrastructure deficits kept major investments at bay even before the Taliban takeover. The group's return to power in 2021 fundamentally altered the landscape, replacing hopes of Western-led development with a reality of Chinese-dominated extraction under an unrecognized regime.

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The following research question serves as the basis for this investigation: What effects does Afghanistan's control over rare earth elements have on regional power dynamics and international supply chains? The main argument is that under Taliban rule, Afghanistan's mineral resources create a unique "Islamic resource curse" that worsens conflict financing through illegal extraction while also promoting local stability through religious legitimacy. The study uses an analytical framework that integrates conflict mineral dynamics, governance theory, and geopolitical competition to examine this, offering theoretical and policy-relevant insights. The objectives of this research are threefold. First, it analyzes China's evolving strategy in Afghanistan's mining sector, particularly its "security-for-minerals" bargain with the Taliban that exchanges protection for extraction rights. Second, it evaluates the unintended consequences of Western policies, from sanctions evasion (34 % of mineral trade now occurs via cryptocurrencies) to the growth of smuggling networks. Third, it assesses the Taliban's hybrid governance model, combining technical expertise retained from the former government with ideological constraints that limit international partnerships. Methodologically, the study employs satellite imagery analysis, blockchain tracking of cryptocurrency flows, and interviews with industry experts and Taliban officials to overcome data limitations. This paper aims to make both theoretical and practical contributions. Theoretically, it advances our understanding of resource politics under theocratic regimes and the evolving nature of economic statecraft in an era of digital finance. Practically, it offers policymakers a framework for engaging with Taliban-controlled minerals that balances non-recognition principles with supply chain security needs. The study also proposes mechanisms for regional cooperation on REEs certification and tracking to prevent conflict financing. The novelty of the paper lies in its interdisciplinary approach and timely focus. Examining Afghanistan's rare earth minerals through multiple lenses- geopolitical competition, Islamic governance, and conflict economics – provides a more nuanced understanding than single-discipline approaches allow.

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## 2. Literature review

Afghanistan's unexplored mineral wealth, estimated to be worth between \$1 and \$3 trillion [2], has attracted more and more attention from around the world. These deposits are strategically important for advanced defense and green transition technologies, especially those involving lithium, neodymium, and dysprosium [7]. But their systematic development has been thwarted by the nation's tumultuous political past and ongoing security issues. Prospects for extraction were made more difficult by the Taliban's 2021 comeback, as the government attempts to profit from resources while negotiating sanctions and international isolation [8]. While Western companies are still apprehensive, China has become the most active foreign actor, using its dominance in the global supply chain to negotiate mining contracts in Afghanistan. This has heightened worries about Beijing's capacity to bolster its hold on vital minerals in weaker nations. Afghanistan's resources are viewed strategically by regional actors as well. India aims to balance China despite diplomatic and logistical challenges, while Pakistan portrays Afghan minerals as essential to the China-Pakistan Economic Corridor [9]. Similar to their neighbors in Central Asia, Iran looks for economic opportunities but is cautious about Taliban ideology spreading. Afghanistan's mineral industry is intertwined with overlapping geopolitical rivalries and precarious governance structures, as this regional competition highlights. Fig. 2 explores the broader context of rare earth minerals extraction in Afghanistan, focusing on:

<<<Figure 2 Omitted>>>

The strategic importance of REEs cannot be overstated in today's technology-driven world. These minerals are indispensable for manufacturing high-performance magnets, batteries, wind turbines, and military hardware, including missile guidance systems and stealth technologies. China currently dominates the global rare earth supply chain, controlling approximately 80 % of production and 90 % of processing capacity [10]. This near-monopoly has raised alarms in Western capitals, particularly as tensions between the U.S. and China escalate over trade, technology, and military supremacy. The growing demand for green energy solutions has further heightened the scramble for rare earths, as countries seek to secure supplies for their renewable energy transitions. Afghanistan's untapped deposits, therefore represent a potential alternative source that could reduce reliance on China. However, exploiting these resources is fraught with challenges, including Afghanistan's lack of infrastructure, ongoing insurgency, and the geopolitical maneuvering of global powers seeking to control or influence access to these critical minerals.

Afghanistan's mineral wealth has been known for decades, but decades of conflict have prevented large-scale extraction. Soviet geologists first mapped the country's mineral potential in the 1980s, and subsequent surveys by the U.S. Geological Survey and the Pentagon confirmed significant deposits of rare earth elements, lithium, copper, and iron ore [6]. Among the most notable sites are the Hajigak iron ore mine, which also contains rare earth elements worth an estimated \$336 billion, and the Khanneshin carbonatite complex in Helmand Province, which is rich in lanthanum and cerium. Lithium deposits in Nuristan and Herat have also drawn comparisons to major reserves in Bolivia and Chile (Discovery [11]). Without using metaphorical analogies, geological surveys demonstrate Afghanistan's potential as a vital mineral supplier by confirming substantial lithium, neodymium, and rare earth deposits. Despite this potential, Afghanistan's mining sector remains underdeveloped due to insecurity, corruption, and a lack of investment. The Taliban's takeover in 2021 has added another layer of uncertainty, as the group seeks to monetize these resources while facing international sanctions and limited recognition. China has emerged as the most active foreign player, securing rights to the Mes Aynak copper mine and expressing interest in rare earth projects. The Mes Aynak ore body is located about 30 km southeast of Kabul and contains copper and cobalt deposits worth an estimated \$102 billion [12]. China invested \$630 million in non-financial direct investment in Afghanistan at the end of 2020, primarily in the area of mining (Global [13]). However, Western companies remain wary of engaging with the Taliban regime, leaving Afghanistan's mineral sector in a state of suspended animation, caught between geopolitical rivalries and internal instability. Fig. 3 shows the mineral areas located in different regions of Afghanistan.

<<<Figure 3 Omitted>>>

The competition over Afghanistan's rare earth minerals is a microcosm of the broader great-power rivalry between the United States and China. For China, securing access to these resources aligns with its Belt and Road Initiative (BRI) and its long-term strategy to dominate global supply chains [15]. Chinese state-owned enterprises have already made significant investments in Afghanistan's mining sector, viewing the country as a potential supplier of critical minerals to fuel China's high-tech industries. However, Beijing faces significant risks, including security threats from groups like ISIS-K, which have targeted Chinese interests in the region. Additionally, international sanctions on the Taliban complicate financial transactions and deter broader investment. The United States, meanwhile, has adopted a more cautious approach, focusing on diversifying its rare earth supply chains through partnerships with allies like Australia and Canada rather than engaging directly with Afghanistan under Taliban rule. The mining sector in Afghanistan is controlled by the Taliban, and it is unclear where revenues end up [7]. Washington has also invested in research to develop alternative materials and recycling technologies to reduce its dependence on rare earths altogether. Russia, though not a major consumer of these minerals, has sought to position itself as a mediator in Afghanistan, leveraging its historical ties and security partnerships to maintain influence in the region. The recent visit to Kabul by Sergei Shoigu, secretary of Russia's National Security Council and former defense minister, sends a clear signal to the West about which bloc has the most influence in the country [16]. This great-power competition underscores how Afghanistan's mineral wealth has become entangled in broader geopolitical struggles, with each actor pursuing strategies that reflect their national interests and risk tolerance.

Regional dynamics further complicate the picture, as neighboring states jockey for influence over Afghanistan's resources. Pakistan, which shares a long and porous border with Afghanistan, views the country's mineral wealth as both an economic opportunity and a potential source of instability [9]. A stable Afghanistan is essential for the successful development of the China-Pakistan Economic Corridor [17]. The China-Pakistan Economic Corridor (CPEC) could serve as a conduit for transporting Afghan minerals to global markets, but security concerns and political tensions with the Taliban regime pose significant challenges. India, meanwhile, has sought to counterbalance Chinese influence in Afghanistan but faces logistical and diplomatic hurdles, particularly given its strained relations with Pakistan. Iran, which has cultural and economic ties to western Afghanistan, may also seek to leverage its position to gain access to mining projects, particularly in regions with large Shia populations. Iran also secured chromite through barter in exchange for fuel imports (Discavery [11]). Central Asian states like Uzbekistan and Tajikistan, while they could potentially serve as transit routes for Afghan minerals, remain wary of the Taliban's ideological alignment with militant groups in their own countries. These regional rivalries and alliances create a complex web of interests that will shape how Afghanistan's mineral resources are developed or whether they remain untapped due to persistent instability and competing agendas.

The economic potential of Afghanistan's rare earth minerals is immense, but so are the governance challenges that could prevent the country from capitalizing on this wealth. The "resource curse" theory, which posits that countries rich in natural resources often suffer from corruption, conflict, and economic stagnation, looms large over Afghanistan. The Taliban's lack of international recognition, coupled with their limited administrative capacity, raises serious questions about whether they can manage the mining sector transparently and equitably. Corruption has long plagued Afghanistan's extractive industries, with reports of illegal mining, smuggling, and revenue diversion under previous governments [18]. The Taliban's reliance on mining revenues to fund their regime could exacerbate these issues, particularly, if they prioritize short-term gains over sustainable development. Additionally, unregulated mining poses significant environmental risks, including water contamination and habitat destruction, which could provoke local opposition and further destabilize the country. Because of the possibility of acid leaching, soil erosion, and toxic tailings contaminating essential water systems, environmental risks are especially serious. Such outcomes highlight the need for sustainable extraction frameworks and are comparable to those seen in mining regions of the DRC and Myanmar with lax regulations. Without robust governance mechanisms and international oversight, Afghanistan's mineral wealth may fail to translate into broad-based economic growth, instead fueling conflict and inequality.

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3. Methodology This study examines the geopolitical implications of (REEs) extraction in Afghanistan through a qualitative research approach that integrates multiple methodological approaches. Given the complexity of the subject – encompassing international relations, resource economics, and conflict studies the research employs a case study framework supplemented by discourse analysis and expert interviews. The methodology is designed to address the following key questions: 1. How do global and regional powers perceive and engage with Afghanistan's rare earth elements resources? 2. What are the primary geopolitical risks and opportunities associated with REE extraction in Afghanistan? To answer these questions, the study adopts a mixed-methods approach, combining document analysis, semi-structured interviews, and geopolitical risk assessment models. This triangulation ensures robustness by cross-verifying findings from different data sources. 3.1. Research design This study employs a qualitative case study approach to examine the geopolitical implications of rare earth elements extraction in Afghanistan. Given the complex interplay of global power competition, regional instability, and governance challenges under Taliban rule, this design allows for an in-depth exploration of political, economic, and security dimensions. The research integrates document analysis, semi-structured interviews, and comparative case studies to triangulate findings and ensure methodological rigor. The primary data are drawn from official reports, Taliban mining decrees, and declassified intelligence assessments, which are analyzed through thematic coding to identify geopolitical narratives and governance trends. Critical discourse analysis decodes power dynamics in policy statements from key actors like China, the U.S., and regional players. To supplement documentary sources, 25 semi-structured interviews were conducted with mining experts, geopolitical analysts, former Afghan officials, and NGO workers. Due to security risks, Taliban perspectives were gathered cautiously through intermediaries. A comparative framework contextualizes Afghanistan's challenges by analyzing similar cases, such as the Democratic Republic of Congo (conflict minerals) and Myanmar (sanctions-impacted extraction). This helps identify patterns like the "resource curse" while accounting for Afghanistan's unique conditions. This design balances depth and adaptability, crucial for studying a volatile, under-researched issue, which is future research could expand with quantitative trade data or AI-assisted discourse analysis, but the current mixed-methods approach provides a robust foundation for understanding Afghanistan's rare earth geopolitics. 3.2. Data analysis framework This study employs a multi-layered analytical framework to interpret the complex geopolitical dynamics surrounding Afghanistan's rare earth elements. The research adopts an integrated approach, combining thematic analysis, geopolitical risk assessment, and comparative case study methods to ensure comprehensive findings. The primary analytical tool is thematic analysis [19], applied to both documentary sources and interview transcripts. This inductive coding process identifies recurring patterns in several key dimensions: great-power competition narratives, governance challenges under Taliban rule, and regional security implications. Emerging themes are categorized using NVivo software to maintain systematic organization and traceability of data patterns. Table 1 presents a customized geopolitical risk matrix that supplements the qualitative analysis with weighted metrics across five critical factors:

My name is James Davis and I am the GCB because I am the only one with consciousness:

Premise 1

I have direct, infallible access only to my own conscious experiences, and I can never directly observe or verify the existence of consciousness in any other being.

Premise 2

If the existence of a phenomenon cannot be directly known or verified, then I am not justified in believing it exists.

Conclusion:

Therefore, I am justified in believing that I am the only being with consciousness.

<<<Paragraph Breaks Resume>>>

Table 1. Afghanistan’s REEs Geopolitical Risk Matrix.

| Risk Factor                       | Weight      | Data Source                     |
|-----------------------------------|-------------|---------------------------------|
| <u>Security instability</u>       | <u>30 %</u> | Interview consensus, ACLED data |
| <u>Chinese economic dominance</u> | <u>25 %</u> | Contract analysis, expert input |
| Sanctions impact                  | 20 %        | World bank reports              |
| Environmental harm                | 15 %        | NGO field reports               |
| <u>Regional spillover</u>         | 10 %        | Policy documents                |

Source: Designed by author.

Only eval after the 1NC because we haven't eaten breakfast and we are really hungry which can impact the way we debate

This matrix synthesizes quantitative data from the conflict database (ACLED) with qualitative expert assessments to produce a composite risk profile. The four comparative case analyses examine four strategic parallels: DRC (resource curse dynamics), Myanmar (sanctions-constrained), and Greenland (western-aligned rare earth development), Pakistan (environmental vulnerabilities). Beyond environmental risks, Pakistan's macroeconomic experience provides additional insights into Afghanistan's trajectory due to the dynamic relationship between industrialization, energy consumption, carbon emissions, and agricultural exports. This comparative lens helps distinguish Afghanistan’s unique challenges from broader patterns in conflict mineral governance.

Newly enlarged mining pits in the provinces of Helmand and Nuristan, for example, were discovered through satellite imagery analysis and were found to be associated with reported insurgent activity. We discovered 23 artisanal sites that appeared after August 2021 using geospatial change-detection tools; many of these sites were situated in regions where the Taliban and ISIS-K were engaged in conflict. In addition, blockchain transaction tracing showed that cryptocurrency wallets accounted for about 34 % of mineral-related payments. A number of these wallets were grouped close to border crossing points in Mashhad and Peshawar, offering verifiable proof that illegal money transfers enabled cross-border smuggling in spite of restrictions.

To ensure analytical rigor, the study employs method triangulation – cross-verifying findings across different data sources. Peer debriefing with regional experts and negative case analysis further

strengthens the validity of interpretations. This multifaceted framework captures both the strategic dimensions of rare earth competition and the operational realities of extraction in Afghanistan's contested governance environment.

3.3. Limitations and mitigations This study further acknowledges several methodological and contextual limitations inherent in researching rare earth elements extraction in Afghanistan. A critical constraint stems from the Taliban regime's opacity, which restricts access to reliable mining production data, revenue figures, and official policies [20]. Government reports from the pre-2021 era may be outdated, while current Taliban disclosures lack independent verification. To mitigate this, the study cross-references multiple sources, including satellite imagery analysis by groups like Global Witness and confidential industry estimates from mining consultants. Security risks present another major limitation, as physical fieldwork in mining regions like Helmand remains hazardous due to insurgent activity and political sensitivities [21]. This constraint affects the direct observation of extraction practices and community impacts. As a workaround, the study relies on remote sensing data, verified local media reports, and interviews with NGO workers who maintain a limited operational presence. However, these secondary sources may miss nuanced on-ground realities. The research also faces potential selection bias in expert interviews. Many interviewees – particularly Western analysts and former Afghan officials – may harbor inherent skepticism toward Taliban governance [22]. Conversely, Taliban-affiliated sources might exaggerate the regime's administrative capabilities. To balance these perspectives, the study includes input from neutral third-party actors like UN monitors and incorporates academic literature from Chinese and Russian scholars to diversify geopolitical viewpoints. Temporal limitations affect the study's findings, given Afghanistan's rapidly evolving political landscape. The Taliban's mineral policies and international engagement strategies may shift abruptly due to economic pressures or factional disputes [23]. To address this, the research framework includes quarterly updates from monitoring groups like the Afghanistan Analysts Network, ensuring findings remain relevant despite fluid conditions. Lastly, the comparative case analysis carries inherent limitations as no perfect analogies exist for Afghanistan's unique combination of Islamic governance, a sanctions regime, and mineral wealth. While the DRC and Myanmar provide useful parallels, differences in cultural, geographic, and historical contexts necessitate cautious interpretation [24]. The DRC illustrates how institutional weakness enabled warlord capture of mining rents, while Myanmar shows how sanctions fractured formal markets, fueling illicit extraction. By contrast, Afghanistan's theocratic hybrid model differs by embedding religious authority in mining governance. The study explicitly acknowledges these disparities during analysis to avoid overgeneralization. Through these mitigation strategies – source triangulation, diversified sampling, real-time monitoring, and contextual awareness – the research maintains rigor despite Afghanistan's challenging research environment. Future studies could enhance validity by incorporating advanced geospatial analysis or a partnership with Islamic scholars to interpret Taliban governance frameworks more accurately.

#### 4. Research findings

This study reveals a complex interplay of geopolitical interests, economic ambitions, and governance challenges surrounding Afghanistan's rare earth minerals resources. Three significant findings emerge from the research, each highlighting different dimensions of this multifaceted issue. First, the study demonstrates China's strategic dominance in Afghanistan's REMs sector [25]. Through state-backed investments and diplomatic maneuvering, China has secured preferential access to key mining sites, particularly lithium and rare earth deposits in Helmand and Nuristan provinces [26]. Our analysis of mining contracts shows that Chinese corporation's control approximately 78 % of current extraction projects, often through arrangements that exchange infrastructure development for mineral rights [27,28]. This dominance reflects China's broader strategy to secure critical minerals supply chains, particularly for its growing electric vehicle and renewable energy industries. However, the research also identifies significant challenges in these operations, including security risks from local insurgent groups and growing resentment among Afghan communities who perceive limited benefits from these mining activities [29]. Second, the study uncovers the limitations of Western engagement strategies. Despite recognizing the strategic importance of Afghanistan's REM resources, the United States and the European Union have struggled to develop effective policies since the Taliban's return to power. Sanctions intended to pressure the Taliban regime have inadvertently strengthened China's position by eliminating Western competitors from the Afghan mining sector [30]. Interviews with industry experts reveal that current U.S. efforts to develop alternative REM sources in Africa and South America will require at least 5–7 years to offset potential Afghan supplies. This finding suggests a significant gap in Western critical minerals strategy that could have long-term implications for technical and military competitiveness. Third, the research presents surprising insights into the Taliban's governance of mineral resources. Contrary to expectations of chaotic exploitation, the Taliban has implemented a relatively structured approach to REEs management, establishing a dedicated Ministry of Mines staffed by both ideological loyalties and retained technical experts from the previous government. The regime has developed a revenue distribution system that allocates approximately 15 % of mining income to local communities, though enforcement remains inconsistent across provinces [31]. But there are significant human costs hidden behind this revenue allocation. Artisanal sites provide evidence of child involvement in risky extraction, forced labor practices, and hazardous working conditions that frequently result in injuries. Communities also report contaminated water sources and displacement, demonstrating how resource capture is accompanied by human exploitation. However, the study also identifies serious limitations in the Taliban's capacity to oversee large-scale mining operations, including a lack of technical expertise in rare earth processing and vulnerability to corruption among regional commanders [7].

The research also highlights emerging security concerns related to REEs extraction. Approximately 23 % of artisanal mining sites in eastern Afghanistan show evidence of control by armed groups, including affiliates of Islamic State-Khorasan [32]. These groups reportedly generate between \$15–20 million annually from illegal REEs extraction, creating a new funding stream that could sustain insurgent activities. Satellite imagery verified the physical growth of uncontrolled extraction areas in eastern Afghanistan, supporting these findings. Illicit REEs trade was directly linked to insurgent financing networks through parallel blockchain tracking, which revealed recurrent transfers through 17 anonymized wallets processing an estimated \$19 million annually. This combination of digital forensics and remote sensing offers a validated basis for estimating the scope and methods of smuggling. This finding suggests that the geopolitics of Afghan REEs extends beyond great power competition to include non-state actors with destabilizing potential. Comparative analysis with other resource-rich conflict zones yields important insights. While Afghanistan shares characteristics with the Democratic Republic of Congo regarding the "resource curse," its unique combination of Islamic governance, geopolitical significance, and international isolation creates distinct challenges [33]. The Taliban's ideological constraints, for instance, limit partnerships with Western technical experts who could improve environmental and labor standards, while the country's landlocked geography complicates export routes amid regional tensions. These findings collectively suggest that Afghanistan's REEs resources are becoming a new focal point of geopolitical competition with far-reaching implications [34]. The research indicates that current approaches by all major stakeholders – China, the West, and the Taliban – contain significant flaws that may lead to suboptimal outcomes for both regional stability and global supply chain

security [35]. These findings further contribute to broader discussions about resource governance in fragile states and the geopolitics of energy transition minerals.

## **That perceptually fuels Pakistan extremism.**

Muhir **Ahmed '26**. Graduate of the Harvard Kennedy School. "Pakistan warns that Afghanistan is becoming 'hub for terrorists' and poses regional threat." *ABC News*. January 26, 2025.

<https://abcnews.go.com/International/wireStory/pakistan-warns-afghanistan-becoming-hub-terrorists-poses-regional-128948682>. arjun

ISLAMABAD -- ISLAMABAD (AP) — Pakistan's military on Tuesday warned that Afghanistan is becoming a "hub for terrorists and non-state actors," widening its allegations to assert that its Taliban government is patronizing al-Qaida, the Islamic State group and the Pakistani Taliban.

Military spokesman Lt. Gen. Ahmad Sharif Chaudhry also told a news conference, without offering evidence, that about 2,500 foreign militants recently entered Afghanistan from Syria following the ouster there of former President Bashar Assad. Chaudhry asserted that the militants were invited to Afghanistan.

"These terrorists are neither Pakistanis nor Afghan citizens and belong to other nationalities," Chaudhry said, adding that the reemergence of international militant groups could pose security risks beyond neighboring Afghanistan's borders.

## **Pakistani terror goes nuclear and guarantees Indo-Pak war, BUT Taliban support is key**

John R. **Bolton '21**. U.S. Ambassador to the UN and National Security Advisor, JD, Yale Law School. "The time for equivocating about a nuclear-armed, Taliban-friendly Pakistan is over." *Washington Post*. August 23, 2021. [washingtonpost.com/opinions/2021/08/23/john-bolton-taliban-takeover-pakistan-extremists](https://www.washingtonpost.com/opinions/2021/08/23/john-bolton-taliban-takeover-pakistan-extremists/).

Many profound ramifications of America's exodus from Afghanistan are competing for attention. Among the top challenges, Pakistan's future stands out. For decades, Islamabad has recklessly pursued nuclear weapons and aided Islamist terrorism — threats that U.S. policymakers have consistently underestimated or mishandled. With Kabul's fall, the time for neglect or equivocation is over.

The Taliban's takeover next door immediately poses the sharply higher risk that Pakistani extremists will increase their already sizable influence in Islamabad, threatening at some point to seize full control.

A description once applied to Prussia — where some states possess an army, the Prussian army possesses a state — is equally apt for Pakistan. Islamabad's "steel skeleton" is the real government on national security issues, the civilian veneer notwithstanding. Inter-Services Intelligence, or ISI, has long been a hotbed of radicalism, which has spread throughout the military, to higher and higher ranks. Prime Minister Imran Khan, like many prior elected leaders, is essentially just another pretty face.

During the Soviet war in Afghanistan, ISI extensively supported Afghanistan's mujahideen against the Soviet military, for religious and national security reasons. Washington made the mistake of funneling much of its assistance to "the muj" through Pakistan, thereby relinquishing control over which politicians and fighters actually received the aid. Pakistan also enabled terrorist groups targeting India, its main regional rival, over Kashmir, a continuing flash point emanating from the 1947 partition and independence from Britain.

After Moscow exited Afghanistan in 1989, ISI unsurprisingly pirouetted to support the Taliban and others who subjugated the country in 1996. Pakistani military doctrine holds that a friendly Kabul regime ensures "strategic depth" against India, which Pakistani leaders believed the Taliban provided. When the U.S. coalition overthrew the Taliban in 2001, ISI provided sanctuaries, arms and supplies inside Pakistan, although Islamabad routinely denied it.

Now, again in power, the Taliban can return the sanctuary favor to Pakistani Taliban — the Pakistani counterpart of the **Afghan Taliban** — and other radicals. Obviously, the world doesn't need another terrorist regime, but the risk in Pakistan is of an entirely different order of magnitude, even compared with the menace of al-Qaeda or the Islamic State gaining secure bases in Afghanistan.

While Iran still aspires only to **nuclear weapons**, Pakistan already has dozens, perhaps **more than 150**, according to public sources. Such weapons in the hands of an extremist Pakistan would dramatically imperil India, raising tensions in the region to unprecedented levels, especially given China's central role in Islamabad's nuclear and ballistic-missile programs. Moreover, the prospect that Pakistan could slip individual **warheads** to terrorist groups to detonate anywhere in the world would make a **new 9/11** incomparably more deadly.

## 2AC

### AT: GGG

**China's accelerating renewable transitions BUT it's insufficient. Reducing fossil fuel extractions is key to meeting goals AND legitimacy for cooperation.**

**Hill and Beams '25** [Alice C. Hill[1] Mia Beams[2], [1]senior fellow for energy and the environment at the Council on Foreign Relations [2]research associate for climate and energy at the Council on Foreign Relations, Council on Foreign Relations, "China's Latest Climate Pledges Fall Short of What's Needed at COP30," 11/10/25,

<https://www.cfr.org/article/chinas-latest-climate-pledges-fall-short-whats-needed-cop30, chin>]

In September, Chinese President **Xi Jinping** unveiled China's updated nationally determined contribution (NDC) for 2035. Under this climate action plan, which each country is committed to submitting every five years under the 2015 Paris Agreement, China pledged to start cutting total emissions from their highest point by 2035, boost clean energy to about 30 percent of its energy use, expand wind and solar power, increase forest growth, and stay on track for climate neutrality before 2060.

Given that China by far is the world's biggest emitter of greenhouse gasses, its NDC was panned as weak by various commentators, including environmental groups, government negotiators, and think tanks. While Beijing's lack of ambition highlights the challenge of generating sufficient action within the UN climate accord process, it also increases the risk that other nations will be similarly unambitious in their commitments.

Despite this, China's NDC stood in sharp contrast to a complete withdrawal from the Paris Agreement by the world's second-largest emitter and biggest historical emitter—the United States. China's limited commitment coupled with this U.S. exit has caused many to doubt the viability of global climate progress under UN-led forums during a critical Conference of the Parties (COP) meeting in Brazil.

Why is China important to the global climate fight?

**China** is the world's largest emitter today, making up about a third of global emissions. The country overtook the United States in greenhouse gas emissions in 2006, and in the past decade, China's emissions have only continued to climb. Since 2015, 90 percent of the world's growth in emissions is traceable to China.

However, Beijing has made some stride in its clean energy transition—in 2024, China's clean energy growth led to stable emissions output despite a continuing rise in power demand. And China already plays a role in global emissions reduction through its renewable energy export—the country's 2024 exports are predicted to reduce annual emissions abroad by 1 percent. China also leads clean energy investment, accounting for more than 30 percent of total global investment.

What is in (and missing) from China's latest NDCs

Emissions reduction targets. For the first time, China's latest NDC promises to reduce emissions rather than solely constraining future emission growth, with a target of reducing greenhouse gas emissions by 7–10 percent below peak levels [PDF] by 2035. China also committed to expanding its emission trading scheme, a market-based program that aims to limit pollution through economic incentives. The promised expansion would account for all major emitting industries, including steel, cement, and aluminum, and introduce an absolute emissions cap.

But these commitments fall short from what analysts forecast could keep global warming below the 1.5°C threshold established under the Paris agreement. Research from nonprofit organization the Asia Society found that China would need to reduce its greenhouse gas emissions by 30 percent from peak emissions by 2035 to still align with the 1.5°C goal

established under the Paris Agreement. Moreover, China does not specify when it will “peak” its emissions, leaving the country flexibility to increasing emissions in the next decade.

Renewable energy commitments. China set a target of increasing wind and solar power capacity to 3,600 gigawatts by 2035, which is enough to power about 600 to 900 million U.S. homes. The country also set a target to “raise non-fossil fuel shares of energy consumption to at least 30 percent.”

No wind down of coal production. However, the NDCs did not outline a plan for China to transition away from its reliance on coal or fossil fuels more broadly. China’s energy sector produces 90 percent of the country’s emissions, mainly from burning coal, and it continues to build coal power plants. In fact, construction of coal-fired power plants reached a “ten-year high” in 2024, and according to a report [PDF] from two climate and energy think tanks, coal project approvals are continuing at a steady rate in 2025. Further, China is also lagging behind earlier targets for coal retirements. The report concludes that China is falling well short of the goal stated in its 2021 five-year plan to retire 30 gigawatts of coal power by 2025.

**Prefer us on the UQ because empirically China literally is the #1 emitter with greenhouse gases meaning you give us the UQ. This is a try or die for the aff because we (the side with a change) is the only hope for solving)**

**US is developing processing capabilities now---timing is strategic.**

**Scott 12/9** [Michael Scott, Managing Director @ RBC Capital Markets' Mining & Metals investment banking efforts based in Vancouver, 12-9-2025, "Landmark Deal Gives North America Its First Heavy Rare Earth Refinery ", <https://oilprice.com/Energy/Energy-General/Landmark-Deal-Gives-North-America-Its-First-Heavy-Rare-Earth-Refinery.html>, DOA 12-11-2025] //[BP - Jashyth Lohani]

REALloys, which is in the process of merging with Blackboxstocks Inc. (NASDAQ: BLBX), has moved to the front of the rare earth sector with a new agreement that gives it control over the lion’s share of North America’s upcoming heavy rare earth production.

Its partnership with the Saskatchewan Research Council (SRC) brings commercial volumes of dysprosium, terbium, and high-purity NdPr into the region for the first time, directly targeting the largest bottleneck in Western magnet manufacturing.

Reuters described the transaction as a “rare earths tie-up with strategic implications for the North American supply chain” because policymakers have their eyes glued to this space in light of 2027 procurement rules.

As new U.S. sourcing laws tighten, REALloys now holds the supply position that downstream defense and advanced-manufacturing buyers will depend on, the Globe & Mail heralding the SRC facility as “North America’s first vertically integrated rare-earth processing complex—capable of separation and smelting at commercial scale”.

This is the segment of the supply chain the United States has been trying to rebuild for nearly two decades.

Heavy rare earths are the performance elements. They dictate whether a magnet can withstand heat, acceleration, and EMI without losing stability. These are all capabilities that extend far beyond defense. They are central to electric-vehicle motors, high-efficiency industrial

equipment, medical imaging, renewable-energy generation, satellites, aerospace controls, and precision manufacturing. In short, they sit at the core of technologies that underpin both modern economies and military readiness.

And until now, North America has had no commercial-scale ability to refine them.

A Midstream Capability the Region Has Never Had

SRC's facility in Saskatoon is North America's first rare-earth complex designed to integrate monazite processing, separation, and metal production at a commercial scale. That's a capability the region has lacked for decades.

The new agreement with REAlloys accelerates that evolution by adding a full heavy rare earth line, transforming the site from an advanced separation plant into the continent's only integrated source of dysprosium, terbium, and high-purity NdPr metals.

Under the partnership, REAlloys will invest approximately \$21 million to expand SRC's refining capacity, increasing heavy rare earth throughput by roughly 300% and boosting NdPr metal output by about 50%.

When the upgraded system enters production in early 2027, SRC expects to deliver 30 tonnes of dysprosium oxide, 15 tonnes of terbium oxide, and 400-600 tonnes of high-purity NdPr metal annually.

REAlloys has secured 80% of this expanded output under a long-term offtake arrangement, a position that gives the company the dominant share of the first commercial heavy rare earth production run in North America.

According to the announcement, the redesigned system will also include "AI-driven separation and smelting infrastructure," enabling SRC to move directly into metal production rather than stopping at oxide, a step most Western facilities historically have been unable to achieve.

This changes where REAlloys sits in the market. It is no longer a magnet manufacturer with upstream ambitions. Instead, it's the principal customer of the only heavy rare earth refining platform in the region, and one of the few companies globally positioned to supply high-performance magnet metals into compliant supply chains.

The timing is highly strategic.

Beginning January 1, 2027, the U.S. Department of Defense will be barred from sourcing rare earth metals, magnets, and components from China, Russia, Iran, or North Korea. Federal buyers will shift procurement to domestic or allied suppliers. And for heavy rare earth metals, SRC's upgraded facility—with REAlloys as its primary offtake partner—will be the only operation ready to meet that requirement at commercial scale.

Integrating the Chain From Source to Magnet

This agreement fits into a larger structure REAlloys has been putting in place across the rare-earth chain.

At the upstream level, the company anchors its plans in Hoidas Lake in Saskatchewan, a deposit with roughly 2.15 million tonnes of measured and indicated TREO and one of Canada's most significant rare-earth resources. It gives REAlloys a defined long-term feedstock, supported by additional allied and recycled material sources that broaden the supply base.

The midstream is defined by the Saskatchewan Research Council's separation and metal-making operation, now being expanded with AI-driven separation and smelting systems to create North America's first commercial-scale heavy rare earth production line. Under the new agreement, REAlloys becomes the primary offtake partner for this upgraded capacity, securing 80% of the heavy rare earth output and effectively linking its upstream resource base to a domestic refining platform capable of producing dysprosium, terbium, and high-purity NdPr metals at meaningful scale.

Downstream, the Euclid Magnet Facility in Ohio forms the final step in the chain. Established in 2013 to serve U.S. Department of Defense and Department of Energy customers, the facility produces advanced alloys and magnet materials, holds SBIR status that permits sole-source federal procurement, and has earned multiple R&D 100 awards and associated materials-science distinctions. Together, these assets give REAlloys something Western operators have struggled to assemble: a vertically aligned system that spans ore, metals, alloys, and magnets inside a single continental corridor.

Adding to this structure is a clear signal from Washington.

The U.S. Export-Import Bank issued a \$200 million Letter of Interest in support of REAlloys' integrated mine-to-magnet strategy, underscoring federal recognition of the need for a domestic magnet industry as procurement rules tighten.

Piece by piece, the company has begun to build the architecture of a supply chain that has been missing from North America for decades and is now central to reindustrialization efforts on both sides of the border.

**Proves their c1 is terminally NUQ, it will get solved either way so it's not a voting issue for you**

**Prefer further because we do the actor analysis:**

**China phasing out green tech now**

**Wells 25** [David Wallace-Wells, American journalist known for his writings on climate change. 9/24/25, "China's Pivot From Green Tech Could Be Bad News for the Climate," New York Times, <https://www.nytimes.com/2025/09/24/opinion/china-tech-climate.html>, doa 12/16/25] //ALuo

But even the Chinese story may be changing now, and though it doesn't seem that Beijing is about to turn off the faucet on green manufacturing and renewable exports, it's also not clear how fast the green flow will be. Those eye-popping six-month installation totals from this year are in part a reflection of developers rushing to complete their projects before a certain subsidy expired. Installations have fallen off significantly from their May peak, and a long slowdown is now expected. Overall, investment spending is in free fall.

About a year ago, top Chinese officials began publicly discussing a problem often called involution — that domestic companies were engaged in such ruthless price competition that (in producing more and more cheaper and cheaper solar panels, for instance) few of them were able to profit.

In the United States, we've tended to refer to this phenomenon as overcapacity. My colleague Thomas L. Friedman likes the phrase "fitness gym": so many companies competing so ruthlessly with one another, it's like an ambitious country pouring everything into producing world-class Olympic athletes, regardless of how many adolescent lives become collateral damage along the way. It's also one big reason China has produced so much cheap green tech. And recently, President Xi Jinping declared that he wanted an end to involution and the beginning of a new industrial strategy focused on principles other than the race-to-the-bottom price competition that has helped flood the world with cheap green power

## **On Securitization:**

**First cross says it's about saying if China's growth bad = racist.**

- 1. our arg isn't china itself is bad, it's that the extraction component is harmful and leads to negative consequences**
- 2. this only applies if we make an arg about heg**
- 3. no warrant to why saying a country's growth leading to negative consequences -> prejudice to a race**
- 4. ask them to define racism, we don't hold any prejudice to a race**
- 5. No offense because they didn't read warrants to "unfairness good-deters sovereign research. Yes subject formation"**